

Normal form

$$\boxed{x' = Ax}$$

vector

matrix

vector

Ex:
$$\begin{cases} x_1' = 2x_1 + t^2 x_2 + (4t + e^t) x_4 \\ x_2' = (\sin t) x_2 + (\cos t) x_3 \\ x_3' = x_1 + x_2 + x_3 + x_4 \\ x_4' = 0 \end{cases}$$

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$$

$$x' = \begin{bmatrix} x_1' \\ x_2' \\ x_3' \\ x_4' \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}'$$

$x' = Ax$, where

$$A = \begin{bmatrix} 2 & t^2 & 0 & 4t + e^t \\ 0 & \sin t & \cos t & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Ex: $my'' + ky = 0 \rightarrow \text{normal form}$

$$x_1 = y, \quad x_2 = y'$$

$$\Rightarrow x_2 = x_1', \quad mx_2' + kx_1 = 0$$

$$\Rightarrow \begin{cases} x_1' = x_2 \\ x_2' = -\frac{k}{m}x_1 \end{cases} \Rightarrow x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad A = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & 0 \end{bmatrix}$$

$$\Rightarrow x' = \begin{bmatrix} 0 & 1 \\ -\frac{k}{m} & 0 \end{bmatrix} x$$

$$a_n(t)y^{(n)} + a_{n-1}(t)y^{(n-1)} + \dots + a_1(t)y' + a_0(t)y = 0$$

→ normal form

$$\begin{aligned} X_1 &= y \\ X_2 &= y' \\ X_3 &= y'' \\ &\vdots \\ X_n &= y^{(n-1)} \end{aligned}$$

$$\Rightarrow \begin{cases} X_1' = X_2 \\ X_2' = X_3 \\ X_3' = X_4 \\ \vdots \\ X_{n-1}' = X_n \\ X_n' = y^{(n)} = -\frac{1}{a_n}(a_0 y + a_1 y' + \dots + a_{n-1} y^{(n-1)}) \\ \quad = -\frac{1}{a_n}(a_0 X_1 + a_1 X_2 + \dots + a_{n-1} X_n) \end{cases}$$

Ex:
$$\begin{cases} 2x'' + 6x - 2y = 0 \\ y'' + 2y - 2x = 0 \end{cases} \longrightarrow \text{normal form}$$

$$\begin{aligned} x_1 &= x \\ x_2 &= x' \\ x_3 &= y \\ x_4 &= y' \end{aligned} \quad \begin{cases} x_1' = x_2 \\ x_2' = x'' = \frac{2y - 6x}{2} = y - 3x = x_3 - 3x_1 \\ x_3' = y' = x_4 \\ x_4' = y'' = 2x - 2y = 2x_1 - 2x_3 \end{cases}$$

$$\Rightarrow x' = \begin{bmatrix} 0 & 1 & 0 & 0 \\ -3 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 2 & 0 & -2 & 0 \end{bmatrix} x$$

Review: linear algebraic equations

$$\begin{cases} a_{11}x_1 + \dots + a_{1n}x_n = b_1 \\ a_{21}x_1 + \dots + a_{2n}x_n = b_2 \\ a_{31}x_1 + \dots + a_{3n}x_n = b_3 \\ \vdots \\ a_{n1}x_1 + \dots + a_{nn}x_n = b_n \end{cases}$$

solution

$\Rightarrow$

$x_1, x_2, \dots, x_n$

1° Unique

2° Inf. many

3° no solutions

Gauss-Jordan elimination

Ex:
$$\begin{cases} 2x_1 + 6x_2 + 8x_3 = 16 \\ 4x_1 + 15x_2 + 19x_3 = 38 \\ 2x_1 + \quad \quad + 3x_3 = 6 \end{cases} \Rightarrow \begin{cases} 2x_1 + 6x_2 + 8x_3 = 16 \\ 3x_2 + 3x_3 = 6 \\ -6x_2 - 5x_3 = -10 \end{cases}$$

$(1) \times (-2) + (2), \quad (1) \times (-1) + (3)$

$$\Rightarrow \begin{cases} 2x_1 + 6x_2 + 8x_3 = 16 \\ 3x_2 + 3x_3 = 6 \end{cases} \quad (2) \times (2) + (3)$$

$x_2 = \frac{6 - 3x_3}{3} = 2 - x_3 = 0$

$$\begin{cases} x_1 = 0 \\ x_2 = 0 \\ x_3 = 2 \end{cases}$$

$\leftarrow$ uniq.

$(1) \Rightarrow x_1 = \frac{16 - 8x_3 - 6x_2}{2} = 8 - 4x_3 - 3x_2 = 0$

$$\underline{\text{Ex:}} \quad \begin{cases} 2x_1 + 4x_2 + x_3 = 8 \\ 2x_1 + 4x_2 = 6 \\ -4x_1 - 8x_2 + x_3 = -10 \end{cases} \Rightarrow \begin{cases} 2x_1 + 4x_2 + x_3 = 8 \\ -x_3 = -2 \\ 3x_3 = 6 \end{cases}$$

$$\Rightarrow \begin{cases} x_1 = 3 - 2s \\ x_2 = s \\ x_3 = 2 \end{cases}$$

↑
inf. many
solutions

$$2x_1 + 4x_2 + 2 = 8$$

$$2x_1 + 4x_2 = 6$$

$$x_1 + 2x_2 = 3$$

$$x_1 = 3 - 2x_2$$

s is parameter

Ex:
$$\begin{cases} 2x_1 + x_3 = -1 \\ -3x_1 + x_2 + 4x_3 = 1 \\ -x_1 + x_2 + 5x_3 = 1 \end{cases} \Rightarrow \begin{cases} -x_1 + x_2 + 5x_3 = 1 \\ 2x_2 + 11x_3 = 1 \\ -2x_2 - 11x_3 = -2 \end{cases}$$

$$(3) \times 2 + (1) \quad , \quad (3) \times (-3) + (2)$$

$$\Rightarrow \begin{cases} -x_1 + x_2 + 5x_3 = 1 \\ 2x_2 + 11x_3 = 1 \\ \boxed{0 = -1} \end{cases} \quad (3) + (2) \quad ? \leftarrow \text{no solutions.}$$